

RESEARCH ARTICLE

Active Learning Framework for Improving Knowledge Graph Accuracy

DONGHYUN KIM^{ID}, HYEONGJUN YANG^{ID}, SEOKJU HWANG, KYUHWAN YEOM, MIDAN SHIM^{ID}, AND KYONG-HO LEE^{ID}, (Member, IEEE)

Department of Computer Science, Yonsei University, Seoul 03722, Republic of Korea

Corresponding author: Kyong-Ho Lee (khlee89@yonsei.ac.kr)

This work was supported by Korea Research Institute for Defense Technology Planning and Advancement (KRIT)–Grant funded by the Defense Acquisition Program Administration (DAPA), South Korea (Technology for Automated Generation and Verification of the Conceptual Models of the Mission Space, South Korea) under Grant 16-202-210-025.

ABSTRACT Knowledge graphs are graph-structured data models that provide a robust scheme for representing real-world relational facts with structured triples. The structural and factual information in knowledge graphs are extensively leveraged in various downstream applications. Unfortunately, knowledge graphs often contain incorrect triples due to the automated extraction processes. Therefore, to ensure the reliability and usability of knowledge graphs, it is crucial to identify and rectify these incorrect triples. However, this remains a challenging task, as knowledge graphs are intricate structures encompassing a vast number of triples formed by diverse entities, relations, and their complex interconnections. This paper proposes an effective method to enhance knowledge graph accuracy by introducing an active learning framework. The proposed framework integrates the advantages of machine-based models and human involvement to enable efficient and reliable improvement in knowledge graph accuracy. Additionally, the proposed method includes sampling strategies that consider the relation distribution in knowledge graphs to maximize the effectiveness of the active learning framework. Extensive experimental results demonstrate the effectiveness of the proposed active learning framework and sampling strategies in improving knowledge graph accuracy. Furthermore, this paper provides an exploration of the level of human involvement and a discussion of practical approaches to improve knowledge graph accuracy in real-world scenarios.

INDEX TERMS Active learning, data cleaning, knowledge graph accuracy.

I. INTRODUCTION

Knowledge graphs (KGs) are graph-structured data models that represent real-world relational facts in the form of a triple (*head*, *relation*, *tail*). KGs provide a robust scheme for capturing semantics in real-world knowledge by organizing entities as nodes and relations between them as directed edges. Their expressive capabilities allow KGs to be extensively employed in downstream applications, including recommendation [1], dialogue generation [2], and question answering [3]. Recently, with the remarkable advancements in large language models (LLMs), research on utilizing and integrating KGs to enhance LLMs performance has gained substantial attention [4]. In particular, the structural

and factual knowledge of KGs significantly contributes to mitigating hallucinations in LLMs [5], [6] and enhancing the reliability of their reasoning processes [7], [8].

Undoubtedly, it is imperative to ensure the quality of KGs to harness their powerful potential across various knowledge-driven applications. However, real-world KGs inevitably contain incorrect triples because they are extracted and merged in an automated manner from large-scale heterogeneous data sources [9]. For instance, the early version of NELL, a prominent KG, was reported to have a precision of 74% [10]. These inaccuracies in KGs are not merely isolated issues; they significantly affect the performance of downstream tasks. To alleviate these challenges, various studies have been conducted to evaluate and enhance the quality of KGs. However, existing approaches face notable limitations: (i) they do not directly improve KG accuracy,

The associate editor coordinating the review of this manuscript and approving it for publication was Francisco J. Garcia-Penalvo^{ID}.

and (ii) they reduce KG coverage by removing incorrect triples (§II-A).

In order to improve the accuracy of KG, a practical framework capable of identifying and rectifying incorrect triples is required. Despite this necessity, addressing inaccuracies in KG remains a challenging task. Intuitively, the most accurate and reliable approach involves the participation of human annotators. However, given the scale of real-world KGs, it is impractical and expensive for human annotators to verify all triples. Meanwhile, a machine-based automated approach can quickly assess all triples but tends to have lower accuracy and reliability compared to human perception [11].

Active learning offers a feasible solution that synergizes the strengths of human expertise and machine-based automation. The active learning process typically involves two stages: (i) a sampling stage, where a model selects the most informative data for annotation, and (ii) a labeling stage, where a human oracle annotates the selected data. Building on this approach, we propose a framework to enhance KG accuracy by integrating the advantages of human involvement and machine-based automation. Specifically, during the sampling stage, the model explores a KG to identify the most suspicious triples, while during the labeling stage, the human oracle rectifies these triples. This interactive framework enables the efficient and reliable improvement of KG accuracy.

The proposed framework comprises four stages: (i) Training, (ii) Scoring, (iii) Sampling, and (iv) KG Update. First, during the training stage, the KG embedding (KGE) model is employed to identify incorrect triples within the target KG. To this end, the task objective is redefined so that the model can identify incorrect triples using solely the internal information of the KG, without relying on auxiliary data. Next, in the scoring stage, the trained KGE model computes the plausibility scores for all triples in the KG. After that, the most suspicious triples, i.e., those to be rectified by the human oracle, are selected through the sampling stage. At this time, we introduce two sampling strategies to efficiently improve the KG accuracy within a limited budget by considering a relation distribution in the KG. Finally, the human oracle rectifies the sampled triples, and the KG is updated accordingly. By repeating this process, the accuracy of the KG is progressively improved.

Extensive experimental results on large-scale KG datasets demonstrate the effectiveness of the proposed framework. We evaluate the framework's performance through experiments conducted in both simulated and human-in-the-loop environments. Furthermore, we present a comprehensive discussion that offers insights into practical and effective strategies for enhancing KG accuracy in real-world scenarios. To the best of our knowledge, this is the first study leveraging an active learning framework for improving KG accuracy.

Our contributions are summarized as follows:

- We propose a novel active learning framework that integrates human involvement and an automated model to enhance the accuracy of KGs.

- By redefining the task objective, we enable the model to identify incorrect triples using only the internal information of the KG.
- We introduce sampling strategies that account for the long-tail relation distribution in the KG to rectify incorrect triples effectively.
- Our extensive experiments on large-scale KGs demonstrate the effectiveness of the proposed framework on KG accuracy improvement.

The rest of this paper is organized as follows. Section II provides a comprehensive review of existing research on KG accuracy and the application of active learning techniques in KG. Section III introduces a novel active learning framework designed to improve knowledge graph accuracy, along with efficient sampling strategies that account for relation distribution. Section IV describes the experimental setup details. Sections V and VI present the experimental evaluation results and analyze the performance of the proposed framework. Section VII provides a comprehensive discussion for applying the proposed framework in real-world scenarios and examines some limitations. Finally, Section VIII concludes the study and discusses potential directions for future research.

II. RELATED WORK

A. KNOWLEDGE GRAPH ACCURACY

Real-world KGs, constructed through automated knowledge extraction processes from vast and diverse data sources, inevitably contain incorrect triples. The inaccuracies not only degrade the overall quality of KGs but also significantly hinder the performance of various downstream tasks. Extensive research has emerged in response to these critical challenges, focusing on enhancing the accuracy and reliability of KGs. As one of them, studies on the evaluation of automatically constructed KGs have been conducted. Ojha and Talukdar [12] propose a method that effectively evaluates the accuracy of an entire KG by leveraging coupling constraints to assess a small subset of triples. Furthermore, Gao et al. [13] propose a KG evaluation framework and an efficient sampling method that selects representative triples to minimize human effort while ensuring a statistically reliable assessment of KG accuracy. Qi et al. [14] propose a human-machine collaboration mechanism to evaluate the accuracy of both triples and entities, aiming to reduce the overall human annotation cost. However, the above studies primarily focus on evaluating KG accuracy and do not directly contribute to the improvement of KG accuracy itself.

Meanwhile, research on KG error detection focuses on enhancing KG accuracy by identifying and removing incorrect triples. Jia et al. [15] propose a triple trustworthiness measurement framework that integrates internal and global information. Wang et al. [16] establish a cross-graph representation learning method that leverages an external KG to validate triples in the target KG. Zhang et al. [17] introduce a contrastive error detection framework that

employs multi-view graph augmentation and an error-aware graph encoder to capture information in each view. Dong et al. [18] propose an ensemble learning framework that integrates multiple error detectors and actively updates the learning policy with minimal human annotation. Recently, Liu et al. [19] present an interactive contrastive learning method using textual and structural graph information to detect semantically similar incorrect triples. Regrettably, the underlying assumption in KG error detection is that KG accuracy would be improved by removing identified errors. This approach results in a loss of KG coverage because it overlooks the potential for rectification in incorrect triples.

B. ACTIVE LEARNING

Active learning (AL) is a methodology designed to optimize model performance with fewer training labels by strategically selecting and annotating data samples within a limited budget. A critical component of AL is sample acquisition, which aims to select the most valuable data points based on evaluations of uncertainty and diversity. Uncertainty-based methods estimate the uncertainty of samples to acquire the top-k most informative samples [20], [21]. In contrast, distribution-based methods select representative samples based on the distribution of data points and model features [22], [23]. Furthermore, hybrid methods have been developed to integrate considerations of both uncertainty and diversity [24], [25].

Nowadays, AL is being widely studied across various fields due to its recognized effectiveness. This study, in particular, investigates the application of AL within the context of KGs. Ostapuk et al. [26] propose an AL framework that adopts a Bayesian view on neural link prediction to select the most informative data samples within the KG. Through this approach, they achieve comparable link prediction performance using only a portion of the KG. Pradhan et al. [27] introduce a semi-automatic KG generation system leveraging AL, facilitating entity recognition and relation identification from a large text corpus. Liu et al. [28] utilize an AL to generate seed alignment to obtain a more effective KG entity alignment. They reduce annotation costs using a structure-aware uncertainty sampling strategy and a bachelor recognizer to avoid non-matchable entities. Seo et al. [29] propose a granularity score-based AL method to expand a seed KG schema with new semantic types. They refine the KG schema with minimal effort by querying human annotators for the most uncertain types that need to be expanded. Huang et al. [30] introduce an alignment framework that jointly aligns entities, relations, and classes across different KGs. The integration of AL optimizes the alignment process by iteratively selecting the most informative pairs for human labeling.

Although substantial progress has been made in leveraging AL's potential in KGs, its application for KG accuracy improvement remains an open challenge. Therefore, in this work, we employ an AL framework to improve KG accuracy.

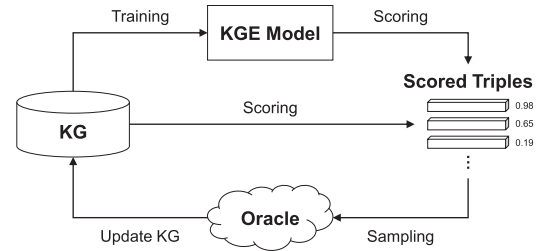


FIGURE 1. Overview of the proposed active learning framework for improving knowledge graph accuracy.

This approach integrates the unique advantages of human involvement and machine-based automation, providing a comprehensive solution to the challenge.

III. METHOD

In this section, we introduce a novel AL framework for improving KG accuracy. Figure 1 presents an overview of the proposed framework. Our framework comprises four stages: (1) Training, where the model is trained on a target KG; (2) Scoring, where the model computes the plausibility scores of triples; (3) Sampling, which selects the most suspicious triples; (4) KG update, which removes incorrect triples and adds the correct ones rectified by the human oracle.

In the remainder of this section, we first define the target problem and describe the proposed AL scenario. Then, we present the triple scoring and sampling strategies for identifying and rectifying incorrect triples.

A. PROBLEM DEFINITION

Let $\mathcal{G}_{initial} = \mathcal{D} \cup \mathcal{D}'$ be an initial KG, which consists of structured triples $(h, r, t) \in \mathcal{E} \times \mathcal{R} \times \mathcal{E}$, where $\mathcal{E}$ is a set of entities, and $\mathcal{R}$ is a set of relations. Here, $\mathcal{D}$ denotes the correct triples, while $\mathcal{D}'$ denotes the incorrect triples that do not align with real-world facts, such as *(BillGates, CEO, Google)*. Our goal is to identify and rectify these incorrect triples through AL iterations in the initial KG $\mathcal{G}_{initial}$, thereby producing a KG $\mathcal{G}_{improve}$ with improved accuracy. Additionally, we aim to maximize the effectiveness of the AL framework, i.e., maximum enhancement in KG accuracy within a limited budget.

B. ACTIVE LEARNING SCENARIO

General AL aims to update the labeled pool by selecting the most informative data sample from the unlabeled pool. In contrast, the objective of our AL framework is to update KG by identifying the most suspicious triples and rectifying them to correct triples. Algorithm 1 describes a detailed procedure of the proposed AL scenario. First, at each iteration k , the model M_k is initialized and trained on $\mathcal{G}_k$, which denotes the KG at the k -th iteration. After training, the model computes the plausibility score of each triple (h, r, t) in the KG using the scoring function. Subsequently, the acquisition function A employs the computed score $\mathcal{X}_k$ to select the most suspicious samples from all triples within $\mathcal{G}_k$.

Then, selected samples $\mathcal{S}$ are provided to the human oracle and rectified to the correct triples $\mathcal{S}_k$. Finally, $\mathcal{G}_k$ is updated to $\mathcal{G}_{k+1}$ by removing $\mathcal{S}$ and adding $\mathcal{S}_k$. Through iterations, we progressively obtain a more accurate KG, ultimately resulting in $\mathcal{G}_{improve}$ with improved accuracy.

Algorithm 1 Active Learning Scenario

Require: Initial KG $\mathcal{G}_{initial}$, Model M , Acquisition function A , Total iterations K

Ensure: Improved KG $\mathcal{G}_{improve}$

```

1: for iterations  $k = 1 : K$  do
2:    $M_k \leftarrow$  Initialize and train on  $\mathcal{G}_k$ 
3:    $\mathcal{X}_k = \emptyset$ 
4:   for triple  $(h, r, t) \in \mathcal{G}_k$  do
5:      $\mathcal{X} \leftarrow M_k(h, r, t)$   $\triangleright$  Compute score
6:      $\mathcal{X}_k = \mathcal{X}_k \cup \{\mathcal{X}\}$ 
7:   end for
8:    $\mathcal{S} \leftarrow A(\mathcal{G}_k, \mathcal{X}_k)$   $\triangleright$  Sample acquisition
9:    $\mathcal{S}_k \leftarrow$  Oracle labels  $\mathcal{S}$ 
10:   $\mathcal{G}_k = \mathcal{G}_k \setminus \mathcal{S}$ 
11:   $\mathcal{G}_k = \mathcal{G}_k \cup \mathcal{S}_k$   $\triangleright$  Update KG
12: end for
13: return  $\mathcal{G}_{improve}$ 

```

C. TRIPLE SCORING

In order to identify incorrect triples, the model is trained to compute the plausibility score of each triple. To this end, we utilize the scoring function $f_r(h, t)$ of the KGE model. However, the target KG contains both correct and incorrect triples, leading the KGE model to operate under the assumption that all triples are valid during the training phase.

The key idea to address this problem is to consider the relativity among plausibility scores of the triples. For example, as illustrated in Figure 2, all triples in the KG are considered valid in the training phase. However, in the inference phase, actual incorrect triples such as *(BillGates, CEO, Google)* tend to have lower plausibility scores compared to correct triples such as *(DonaldTrump, bornin, NewYork)* because they are uncommon patterns within real-world KGs, even if they are classified as true. Inspired by this observation, we frame the incorrect triple identification as a binary classification task to determine whether the triple is true or not. To this end, we design a scoring function to output the probability that a given triple is true. Specifically, we reformulate the original scoring function by introducing a sigmoid layer as follows:

$$s(h, r, t) = \text{sigmoid}(f_r(h, t)). \quad (1)$$

This approach enables the model to efficiently differentiate between correct and incorrect triples based solely on the internal information in the KG. Furthermore, by incorporating a simple superficial layer, the proposed framework ensures compatibility with various KGE models in a model-agnostic manner without extensive modifications.

As previously mentioned, during training, the model operates under the assumption that all triples in the KG

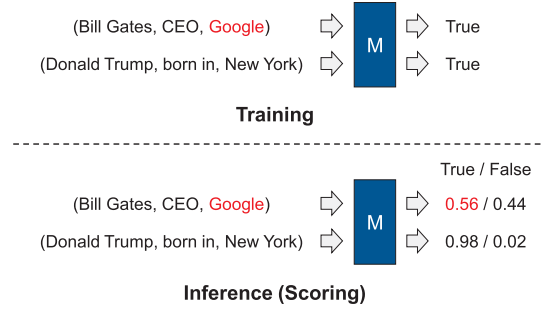


FIGURE 2. Motivating example of identifying incorrect triples with only internal information in KG.

are valid. Nonetheless, leveraging explicit negative triples is essential for training the model to perform binary classification effectively. Therefore, we adopt the negative sampling technique introduced by Bordes et al. [31] to provide the model with explicit negative labels. Negative samples are generated by randomly corrupting either the head or tail entity as follows:

$$\mathcal{G}' = \{(h', r, t) | h' \in \mathcal{E}\} \cup \{(h, r, t') | t' \in \mathcal{E}\}. \quad (2)$$

Note that $\mathcal{G}'$ is a set of synthetic incorrect triples generated by negative sampling for model training, while $\mathcal{D}'$ is the actual incorrect triples within the KG, where $\mathcal{G} = \mathcal{D} \cup \mathcal{D}'$.

Finally, the model is trained with the goal of minimizing the binary cross-entropy loss as follows:

$$-\sum_{\tau \in \mathcal{G}} \sum_{\tau' \in \mathcal{G}'} (y \cdot \log(s(\tau)) + (1 - y) \cdot \log(1 - s(\tau'))), \quad (3)$$

where τ and τ' refer to positive and negative triples, respectively, and $y \in \{1, 0\}$ is the corresponding label.

As described in Algorithm 1, the model is initialized and retrained on the KG at each AL iteration. Conventional AL training methods, such as incremental training, are unsuitable for our framework. This is because incremental training on updated triples affects existing KG embedding, potentially reducing the model's generalization capability. Recently, methods to dynamically adjust only the embedding of the updated triple have been proposed [32], [33]. However, applying such approaches is beyond the scope of this study and is left for future work.

D. TRIPLE SAMPLING

1) RANDOM SAMPLING (RD)

Random sampling is a commonly used and straightforward strategy for selecting samples. As a baseline, we randomly select triples from KG.

$$\mathcal{S} = \text{random}(\psi_1, \dots, \psi_{|\mathcal{G}|}), \quad (4)$$

where $\psi_i = (h, r, t)$ denotes each triple in $\mathcal{G}$.

2) PLAUSIBILITY SAMPLING (PS)

We sample the triples using the plausibility score π calculated by the scoring function. The triples with the lowest scores are

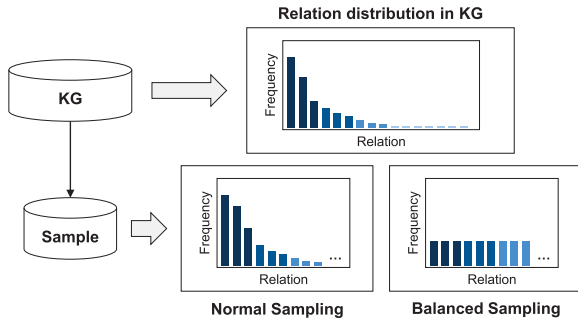


FIGURE 3. Illustration of the distribution differences between normal sampling and balanced sampling.

selected as follows:

$$\pi_i = s(\psi_i), \quad (5)$$

$$S = \operatorname{argmin}(\pi_1, \dots, \pi_{|\mathcal{G}|}). \quad (6)$$

3) UNCERTAINTY SAMPLING (UC)

We adopt uncertainty as the sampling criterion based on the assumption that the model's predictions for incorrect triples will exhibit high uncertainty. The scoring function outputs the plausibility score, representing the probability of a given triple being true. Based on this score, we calculate the entropy η and select triples with the highest entropy as follows:

$$\eta_i = -\pi_i \cdot \log \pi_i - (1 - \pi_i) \cdot \log(1 - \pi_i), \quad (7)$$

$$S = \operatorname{argmax}(\eta_1, \dots, \eta_{|\mathcal{G}|}). \quad (8)$$

4) BALANCED SAMPLING

Relations in real-world KGs follow a long-tailed distribution [34]. Specifically, a small proportion of relations contributes to the majority of the triples in the KG, while a large proportion of relations contains only a few numbers of triples. The normal sampling strategy of selecting triples in order based on their scores fails to address the imbalance of relations. As illustrated in Figure 3, the selected samples may reflect imbalances in relations distribution. Consequently, while the overall accuracy of the KG is increased, the impact of improvement is limited to a few prominent relations and does not extend to all relations. Therefore, we propose a balanced sampling strategy that takes into account the distribution of relations to mitigate the imbalance problem. Balanced sampling selects triples based on their scores but ensures that a uniform number of triples are chosen for each relation. Algorithm 2 describes the balanced sampling process. We apply the balanced sampling to the score-based PS and UC strategies. In line 5, $\operatorname{argsort}(\cdot)$ function is applied in ascending order for the PS and descending order for the UC.

5) CLUSTERED SAMPLING

As another approach to address sample imbalance, we propose a clustered sampling strategy that integrates the distribution of relations within the KG and the semantic information

Algorithm 2 Balanced Sampling

Require: KG $\mathcal{G}$, Computed score $\mathcal{X}$, Number of relations N_r , Sampling size N_s

Ensure: Sampled triples S

```

1:  $S = \emptyset$ 
2:  $N_{pr} \leftarrow N_s / N_r$   $\triangleright$  Number of sample per relation
3:  $\mathcal{X}_r \leftarrow$  Score of triples with relation  $r$ 
4: for each relation  $r$  in  $\mathcal{G}$  do
5:    $\mathcal{B} \leftarrow \operatorname{argsort}(\mathcal{X}_r)[1:N_{pr}]$ 
6:    $S = S \cup \{\mathcal{B}\}$ 
7: end for
8: return  $S$ 
```

between triples. Unlike the balanced sampling strategy, which considers explicit relations, the clustered sampling strategy accounts for the implicit semantics between triples. When the number of triples belonging to a given relation is insufficient, balanced sampling may not adequately resolve the relation imbalance. To address this, we utilize clustering to take account of semantic similarities among triples. If the triples belong to the same cluster, the effect of improvement propagates among them, even if the explicit relations are different. The clustered sampling process is depicted in Algorithm 3. First, we obtain the embeddings of all triples in the KG and cluster them using the k-means++ algorithm [35]. Then, we sample N_{pc} triples from each cluster in order of their scores. Like the balanced sampling, the clustered sampling is applied to both PS and UC strategies.

Algorithm 3 Clustered Sampling

Require: KG $\mathcal{G}$, Computed score $\mathcal{X}$, Number of clusters N_c , Sampling size N_s

Ensure: Sampled triples S

```

1:  $C \leftarrow \text{k-means++}(\mathcal{G}, N_c)$ 
2:  $S = \emptyset$ 
3:  $N_{pc} \leftarrow N_s / N_c$   $\triangleright$  Number of sample per cluster
4:  $\mathcal{X}_c \leftarrow$  Score of triples in cluster  $c$ 
5: for each cluster  $c$  in  $C$  do
6:    $\mathcal{B} \leftarrow \operatorname{argsort}(\mathcal{X}_c)[1:N_{pc}]$ 
7:    $S = S \cup \{\mathcal{B}\}$ 
8: end for
9: return  $S$ 
```

IV. EXPERIMENTAL SETUP

To evaluate the proposed framework, we conducted experiments in two distinct environments: (i) simulated and (ii) human-in-the-loop. In the simulated setup, we automatically emulated the process of triple rectification, the most time-consuming step in the AL framework. This environment primarily focuses on evaluating the efficacy of the proposed framework in improving KG accuracy while also analyzing the impact of different KGE models and sampling strategies on its performance (§V). The human-in-the-loop setup is an environment to explore the impact of human involvement

TABLE 1. Statistics of the KG datasets.

Dataset	# Rel	# Ent	Train	Valid	Test
FB15K237	237	14,541	272,115	17,535	20,466
WikiMovie300K	294	36,001	240,000	23,310	23,373
WikiMovie1M	394	104,500	950,000	18,912	18,984
UMLS	46	135	5,216	652	661

on improving the KG accuracy of the framework. In this environment, triple rectification is performed by human oracles with various conditions (§VI).

A. DATASETS

We evaluated the proposed framework using following KG datasets: FB15K237 [36], WikiMovie [26], and UMLS [37]. FB15K237, a subset of Freebase, is one of the most widely used datasets for evaluating KG-related tasks. WikiMovie, a subset of Wikidata, comprises movie-related facts, such as directors, actors, and genres. WikiMovie provides two versions of the dataset: WikiMovie300K, which includes 294 relations, 36,001 entities, and 286,683 triples and WikiMovie1M, which contains 394 relations, 104,500 entities, and 987,896 triples. We utilized WikiMovie to experiment in a more realistic environment involving a large number of relations and entities. UMLS, the upper ontology of the Unified Medical Language System, categorizes concepts into 135 semantic types and links them with 46 relations in the biomedical domain. UMLS is adopted for human-in-the-loop experiments requiring more expertized domain-specific knowledge. The statistics of datasets are described in Table 1.

We synthetically generated incorrect triples to construct $\mathcal{G}_{initial}$ containing errors for AL experiments. We assume that benchmark KG datasets do not inherently include naturally occurring incorrect triples. Following the previous study [38], given a correct triple (h, r, t) , we corrupted either the head or tail entity to a random other entity in the KG to generate the incorrect triple (h', r, t) or (h, r, t') . To make the generated incorrect triples more realistic, the entities h' or t' are constrained to those that appear as the head or tail of relation r in the KG. Note that the synthetic incorrect triples are not in the original KG. The ratio of incorrect triples is set to 10% and 20%. At that time, we set aside the original KG set without the corrupted triples. This allows us to simulate the AL procedure since we already have the correct triples that need to be obtained by a human oracle. The Valid and Test sets are reserved for downstream task experiments.

B. KG EMBEDDING MODELS

In this study, we employed translational distance-based and semantic matching-based methods as KGE models for our framework. These models are simple and effective, achieving fast computation and competitive performance using only internal information in the KG. We selected TransE [31], TorusE [39], DistMult [40], and ComplEx [41] as our base models. TransE [31] is the most representative translational

distance-based KGE model, projecting entities and relations in vector space and ensuring embeddings follow translation $\mathbf{h} + \mathbf{r} = \mathbf{t}$. TorusE [39] addresses TransE's regularization issue by embedding in an n -dimensional torus space, a compact Lie group, where entities and relations are mapped $x \mapsto [x]$, and embeddings are learned through translation, $[\mathbf{h}] + [\mathbf{r}] = [\mathbf{t}]$. DistMult [40] is a semantic matching-based KGE model that uses a diagonal relation matrix to evaluate triples via the scoring function $\mathbf{h}^\top \cdot \mathbf{M}_r \cdot \mathbf{t}$. ComplEx [41] extends DistMult by using complex-valued embeddings to effectively model asymmetric relations, placing entity and relation embeddings in a complex space instead of a real space. It is worth noting that our model-agnostic framework allows it to incorporate other models based on the user's needs and the available environment. Therefore, we employed an LM-based method PALT [42] to demonstrate the applicability of our framework across state-of-the-art KGE models. PALT leverages transfer learning to optimize pre-trained LM by introducing a parameter-lite encoder that adjusts only a small subset of parameters. This approach avoids fine-tuning all LM parameters and significantly reduces computational overhead, making it well-suited for our AL framework.

C. SAMPLING STRATEGIES

To identify incorrect triples, we employed the following sampling strategies in our AL experiments:

- RD: Randomly sample from all triples within the KG.
- PS: Sampling by plausibility of triples.
- PS-balance: PS with relation-aware balanced sampling.
- PS-cluster: PS with semantic-aware clustered sampling.
- UC: Sampling by uncertainty of triples.
- UC-balance: UC with relation-aware balanced sampling.
- UC-cluster: UC with semantic-aware clustered sampling.

D. EVALUATION METRICS

KG Accuracy (Acc) is defined as the proportion of correct triples within the KG and is used to assess the mean accuracy of triples at each iteration.

$$Acc = \frac{|\mathcal{G}^{correct}|}{|\mathcal{G}|} \quad (9)$$

where $\mathcal{G}^{correct}$ is correct triples in $\mathcal{G}$.

Sampling Efficiency (Eff_{smp}) is utilized to assess the efficiency of each sampling strategy. This metric represents the proportion of sampled triples that are actually incorrect and is calculated as follows:

$$Eff_{smp} = \frac{\sum_{k=1}^K \frac{|\mathcal{S}_k^{incorrect}|}{|\mathcal{S}_k|}}{K} \quad (10)$$

where $\mathcal{S}_k^{incorrect}$ is incorrect triples in selected samples $\mathcal{S}_k$ at k -th iteration. The higher value of Eff_{smp} indicates the more efficient use of human efforts, allowing the limited AL budget to be utilized more effectively.

TABLE 2. Results of KG accuracy improvement on FB15K237 and WikiMovie300K. R denotes the ratio of incorrect triples in the dataset. The sampling size is set to 1,000 for R = 0.1 and 2,000 for R = 0.2. The number of clusters for clustered sampling is set to 500.

		FB15K237				WikiMovie300K			
		R = 0.1		R = 0.2		R = 0.1		R = 0.2	
KGE Model	Sampling	$Acc(\Delta)$	Eff_{smp}	$Acc(\Delta)$	Eff_{smp}	$Acc(\Delta)$	Eff_{smp}	$Acc(\Delta)$	Eff_{smp}
TransE [31]	RD	0.9102($\uparrow$ 0.0102)	0.0929	0.8399($\uparrow$ 0.0399)	0.1811	0.9099($\uparrow$ 0.0099)	0.0953	0.8371($\uparrow$ 0.0371)	0.1783
	PS	0.9819 ($\uparrow$ 0.0819)	0.7429	0.9681 ($\uparrow$ 0.1681)	0.7626	<u>0.9651</u> ($\uparrow$ 0.0651)	<u>0.6252</u>	<u>0.9372</u> ($\uparrow$ 0.1372)	<u>0.6589</u>
	PS-balance	0.9479($\uparrow$ 0.0479)	0.4349	0.9316($\uparrow$ 0.1316)	0.5970	0.9392($\uparrow$ 0.0392)	0.3768	0.8987($\uparrow$ 0.0987)	0.4712
	PS-cluster	<u>0.9817</u> ($\uparrow$ 0.0817)	<u>0.7406</u>	<u>0.9680</u> ($\uparrow$ 0.1680)	<u>0.7621</u>	0.9653 ($\uparrow$ 0.0653)	0.6264	0.9375 ($\uparrow$ 0.1375)	0.6601
	UC	0.80(-)	-	0.80(-)	-	0.80(-)	-	0.80(-)	-
	UC-balance	0.80(-)	-	0.80(-)	-	0.80(-)	-	0.80(-)	-
	UC-cluster	0.80(-)	-	0.80(-)	-	0.80(-)	-	0.80(-)	-
TorusE [39]	RD	0.9101($\uparrow$ 0.0101)	0.0918	0.8393($\uparrow$ 0.0393)	0.1785	0.9097($\uparrow$ 0.0097)	0.0926	0.8377($\uparrow$ 0.0377)	0.1812
	PS	<u>0.9465</u> ($\uparrow$ 0.0465)	<u>0.4220</u>	0.9318 ($\uparrow$ 0.1318)	0.5982	<u>0.9314</u> ($\uparrow$ 0.0314)	<u>0.3014</u>	<u>0.8874</u> ($\uparrow$ 0.0874)	<u>0.4197</u>
	PS-balance	0.9392($\uparrow$ 0.0392)	0.3552	0.9191($\uparrow$ 0.1191)	0.5404	0.9229($\uparrow$ 0.0229)	0.2202	0.8652($\uparrow$ 0.0652)	0.3131
	PS-cluster	0.9469 ($\uparrow$ 0.0469)	0.4251	<u>0.9314</u> ($\uparrow$ 0.1314)	<u>0.5962</u>	0.9315 ($\uparrow$ 0.0315)	0.3021	0.8889 ($\uparrow$ 0.0889)	0.4271
	UC	0.80(-)	-	0.80(-)	-	0.80(-)	-	0.80(-)	-
	UC-balance	0.80(-)	-	0.80(-)	-	0.80(-)	-	0.80(-)	-
	UC-cluster	0.80(-)	-	0.80(-)	-	0.80(-)	-	0.80(-)	-
DistMult [40]	RD	0.9102($\uparrow$ 0.0102)	0.0932	0.8395($\uparrow$ 0.0395)	0.1791	0.9101($\uparrow$ 0.0101)	0.0973	0.8382($\uparrow$ 0.0382)	0.1835
	PS	0.9640($\uparrow$ 0.0640)	0.5807	0.9499($\uparrow$ 0.1499)	0.6800	0.9382($\uparrow$ 0.0382)	0.3664	0.8785($\uparrow$ 0.0785)	0.3772
	PS-balance	0.9470($\uparrow$ 0.0470)	0.4259	0.9327($\uparrow$ 0.1327)	0.6020	0.9335($\uparrow$ 0.0335)	0.3218	0.8720($\uparrow$ 0.0720)	0.3458
	PS-cluster	0.9634($\uparrow$ 0.0634)	0.5751	0.9505($\uparrow$ 0.1505)	0.6826	<u>0.9397</u> ($\uparrow$ 0.0397)	<u>0.3809</u>	<u>0.8802</u> ($\uparrow$ 0.0802)	<u>0.3853</u>
	UC	0.9666($\uparrow$ 0.0666)	0.6044	0.9569($\uparrow$ 0.1569)	0.7117	0.9393($\uparrow$ 0.0393)	0.3774	0.8787($\uparrow$ 0.0787)	0.3780
	UC-balance	0.9468($\uparrow$ 0.0468)	0.4245	0.9335($\uparrow$ 0.1335)	0.6056	0.9334($\uparrow$ 0.0334)	0.3208	0.8727($\uparrow$ 0.0727)	0.3490
	UC-cluster	0.9688 ($\uparrow$ 0.0688)	0.6236	0.9593 ($\uparrow$ 0.1593)	0.7228	0.9400 ($\uparrow$ 0.0400)	0.3844	0.8806 ($\uparrow$ 0.0806)	0.3873
ComplEx [41]	RD	0.9106($\uparrow$ 0.0106)	0.0959	0.8402($\uparrow$ 0.0402)	0.1826	0.9098($\uparrow$ 0.0098)	0.0934	0.8373($\uparrow$ 0.0373)	0.1793
	PS	0.9642($\uparrow$ 0.0642)	0.5825	0.9357($\uparrow$ 0.1357)	0.6156	0.9335($\uparrow$ 0.0335)	0.3218	0.8915($\uparrow$ 0.0915)	0.4395
	PS-balance	0.9471($\uparrow$ 0.0471)	0.4272	0.9283($\uparrow$ 0.1283)	0.5821	0.9291($\uparrow$ 0.0291)	0.2794	0.8797($\uparrow$ 0.0797)	0.3829
	PS-cluster	0.9619($\uparrow$ 0.0619)	0.5611	0.9373($\uparrow$ 0.1373)	0.6231	0.9343($\uparrow$ 0.0343)	0.3289	0.8910($\uparrow$ 0.0910)	0.4372
	UC	0.9696($\uparrow$ 0.0696)	0.6313	0.9414($\uparrow$ 0.1414)	0.6414	0.9370 ($\uparrow$ 0.0370)	0.3552	0.8932 ($\uparrow$ 0.0932)	0.4475
	UC-balance	0.9477($\uparrow$ 0.0477)	0.4323	0.9283($\uparrow$ 0.1283)	0.5822	0.9269($\uparrow$ 0.0269)	0.2583	0.8794($\uparrow$ 0.0794)	0.3812
	UC-cluster	0.9707 ($\uparrow$ 0.0707)	0.6412	0.9437 ($\uparrow$ 0.1437)	0.6520	<u>0.9366</u> ($\uparrow$ 0.0366)	<u>0.3516</u>	<u>0.8923</u> ($\uparrow$ 0.0923)	<u>0.4433</u>
PALT [42]	RD	0.9104($\uparrow$ 0.0104)	0.0943	0.8397($\uparrow$ 0.0397)	0.1802	0.9102($\uparrow$ 0.0102)	0.0979	0.8382($\uparrow$ 0.0382)	0.1835
	PS	0.9651($\uparrow$ 0.0651)	0.5904	0.9551($\uparrow$ 0.1551)	0.7035	0.9403($\uparrow$ 0.0403)	0.3868	0.8942($\uparrow$ 0.0942)	0.4522
	PS-balance	0.9474($\uparrow$ 0.0474)	0.4299	0.9394($\uparrow$ 0.1394)	0.6323	0.9321($\uparrow$ 0.0321)	0.3081	0.8833($\uparrow$ 0.0833)	0.4000
	PS-cluster	0.9645($\uparrow$ 0.0645)	0.5850	0.9564($\uparrow$ 0.1564)	0.7094	0.9447($\uparrow$ 0.0447)	0.4291	0.8950($\uparrow$ 0.0950)	0.4563
	UC	0.9712($\uparrow$ 0.0712)	0.6458	0.9589($\uparrow$ 0.1589)	0.7205	0.9492($\uparrow$ 0.0492)	0.4723	0.8984($\uparrow$ 0.0984)	0.4726
	UC-balance	0.9502($\uparrow$ 0.0502)	0.4553	0.9407($\uparrow$ 0.1407)	0.6383	0.9308($\uparrow$ 0.0308)	0.2956	0.8827($\uparrow$ 0.0827)	0.3971
	UC-cluster	0.9781 ($\uparrow$ 0.0781)	0.7084	0.9650 ($\uparrow$ 0.1650)	0.7482	0.9514 ($\uparrow$ 0.0514)	0.4934	0.9011 ($\uparrow$ 0.1011)	0.4854

We also conducted link prediction experiments to assess the impact of improvement in KG accuracy on the downstream tasks. The performance of link prediction is evaluated using Mean Reciprocal Rank (MRR) and Hits@N metrics. Following Bordes et al. [31], we report results using a filtered setting, which excludes all corrupted triples in the Train, Valid, and Test sets before obtaining the ranked list.

E. IMPLEMENTATION DETAILS

The framework is developed in Python, leveraging the open-source library TorchKGE [43] and the official implementation of PALT.¹ For TransE, TorusE, DistMult, and ComplEx, the embedding dimension and the number of negative samples are set to 200 and 25, respectively. The training objective is optimized using the Adam [44], with a learning rate varying between 1e-6 and 1e-3. Training and

scoring batch sizes are empirically chosen from {2048, 4096, 8192} and {128, 256, 512}, respectively. The model is trained up to 500 epochs, and the output from the last epoch is used for triple scoring. For PALT, a pre-trained BERT-base [45] is employed and optimized with AdamW [46]. The batch sizes for training and scoring are set to 32 and 16, respectively. Training is conducted for up to 30 epochs, with the output from the last epoch utilized for triple scoring. The remaining hyper-parameters follow the original setup. All experiments are conducted on a GeForce RTX 3090 GPU.

V. SIMULATED EXPERIMENTS

This section presents simulated experimental results to evaluate the effectiveness of the proposed AL framework. In the simulated setting, triple rectification is performed automatically using predefined correct answer triples. All reported results represent the average of five independent

¹<https://github.com/yuanyehome/PALT>

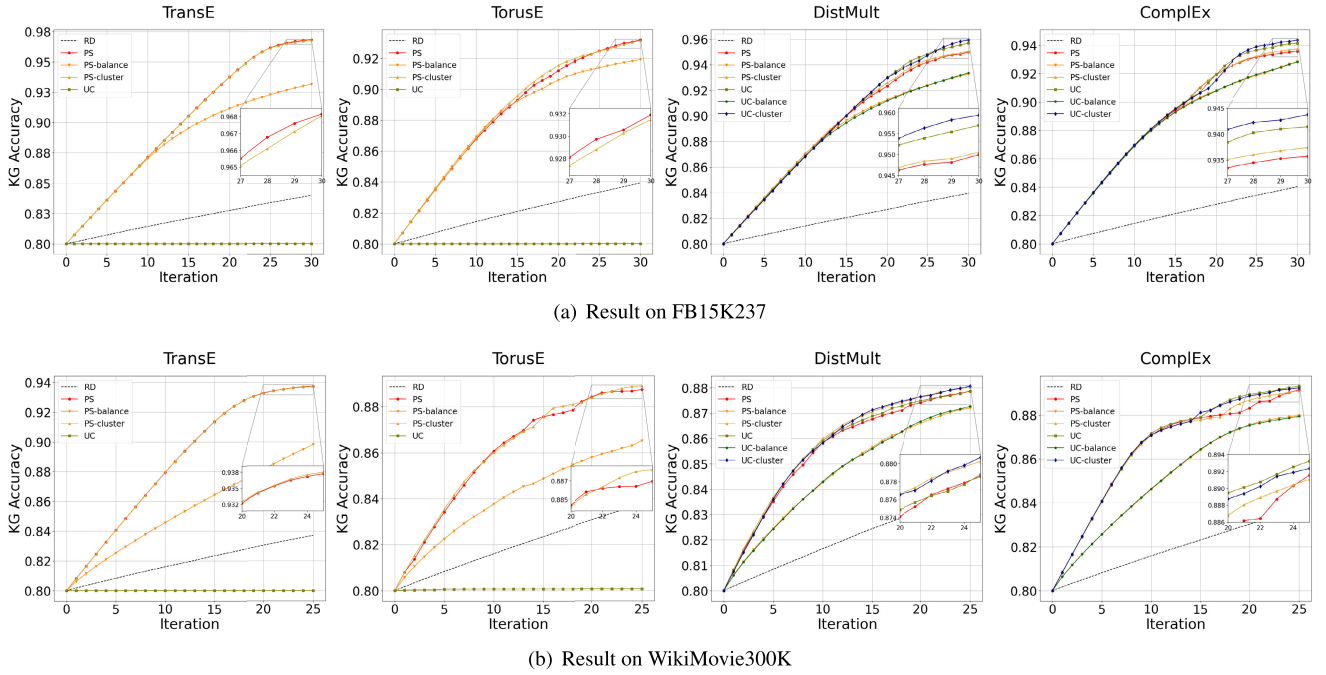


FIGURE 4. KG accuracy at each AL iteration. The sampling size and the number of clusters are set to 2,000 and 500, respectively.

iterations, each with random initialization. The experiments are designed to address the following main research questions (RQs).

- RQ1: How effectively does the proposed AL framework improve KG accuracy?
- RQ2: How does the performance of the framework vary across different KGE models?
- RQ3: How do different AL sampling strategies impact KG accuracy improvement?
- RQ4: How do different AL sampling strategies impact the downstream task performance?

A. RESULTS ON KG ACCURACY IMPROVEMENT (RQ1)

We first investigated the overall performance of the proposed framework. As shown in Table 2, the results demonstrate that our framework significantly enhanced KG accuracy. Specifically, when applied to FB15k237 with an initial KG accuracy of 90% ($R = 0.1$), the accuracy improvements with TransE, TorusE, DistMult, ComplEx, and PALT are up to 8.19% p, 4.69% p, 6.88% p, 7.07% p, and 7.81% p, respectively. Furthermore, with an initial KG accuracy of 80% ($R = 0.2$), TransE, TorusE, DistMult, ComplEx, and PALT demonstrate accuracy improvements of up to 16.81% p, 13.18% p, 15.93% p, 14.37% p, and 16.50% p, respectively.

Our framework also demonstrates significant accuracy improvement on WikiMovie300K, which contains more entities and relations than FB15K237. Specifically, with an initial KG accuracy of 90% ($R = 0.1$), the accuracy enhancements using TransE, TorusE, DistMult, ComplEx, and PALT are up to 6.53% p, 3.15% p, 4.00% p, 3.70% p, and 5.14% p, respectively. Furthermore, when starting from

an initial accuracy of 80% ($R = 0.2$), KG accuracy improvements reach up to 13.75% p, 8.89% p, 8.06% p, 9.32% p, and 10.11% p, respectively. These findings imply that the proposed framework is model-agnostic and successfully enhances the accuracy of KG regardless of its size or structural complexity due to the number of relations and entities.

B. IMPACTS OF KGE MODELS ON KG ACCURACY (RQ2)

We compared the performance of KG accuracy improvement across different KGE models. As shown in Table 2, while all models demonstrate improvements in KG accuracy, TransE notably achieves the highest average accuracy enhancement. Furthermore, TransE shows stable performance across different datasets and error ratios regarding accuracy improvement compared to other models, including PALT, a state-of-the-art LM-based model. These results suggest that KG accuracy improvement is not necessarily dependent on model complexity, underscoring the effectiveness of even simpler models within the proposed framework.

Meanwhile, we observed that incorporating uncertainty (UC) does not improve KG accuracy in translation-based models. Although this result aligns with expectations, it is noteworthy that uncertainty-based sampling strategies have proven ineffective for translation-based models. This ineffectiveness arises from the inherent design of these models, which align embedding through transitions defined by the scoring function, i.e., $\mathbf{h} + \mathbf{r} = \mathbf{t}$, without any trainable parameters capable of measuring the uncertainty of triples. In contrast, semantic matching-based models, such as DistMult, employ a relation matrix $\mathbf{M}_r$ in the scoring function

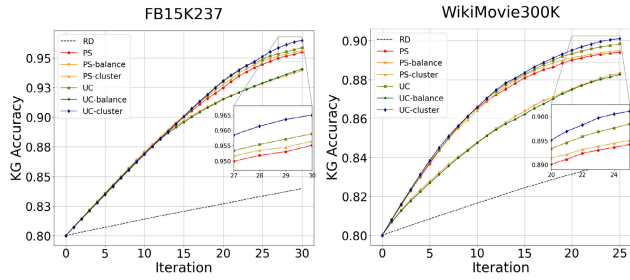


FIGURE 5. KG accuracy at each AL iteration with PALT. The sampling size and the number of clusters are 2,000 and 500, respectively.

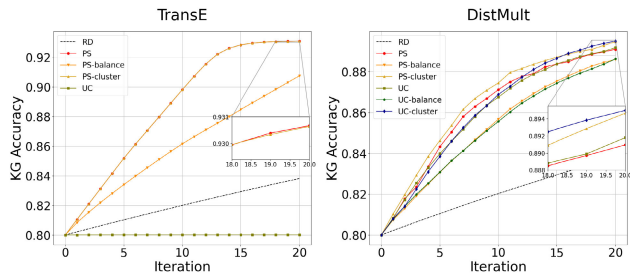


FIGURE 6. KG accuracy at each AL iteration on WikiMovie1M. The sampling size and the number of clusters are 10,000 and 500, respectively.

$\mathbf{h}^T \cdot \mathbf{M}_r \cdot \mathbf{t}$, which works as a kind of trainable layer, enabling uncertainty assessment of model outputs. As a result, these semantic matching-based models can effectively measure the uncertainty of triples.

C. IMPACTS OF SAMPLING STRATEGIES ON KG ACCURACY (RQ3)

We investigated the performance of KG accuracy improvements across different sampling strategies. Figures 4, 5, and 6 illustrate the KG accuracy at each AL iteration for the KG, starting with an initial accuracy of 80% ($R = 0.2$). As anticipated, both PS and UC significantly outperform RD across all KG datasets. PS and UC demonstrate substantial superiority over RD in terms of both accuracy improvement and sampling efficiency, which is at least twice as high compared to RD. These findings indicate that PS and UC not only lead to greater accuracy improvements but also enable more efficient utilization of a limited AL budget. That is, within the same annotation budget, employing PS and UC strategies allows for obtaining a more significant number of correct triples.

In comparing PS and UC, UC yields better KG accuracy improvements and sampling efficiency. As shown in Figures 4 and 5, UC outperforms PS in DistMult, ComplEx, and PALT. Specifically, PS and UC achieve comparable accuracy improvements during the early and middle stages of AL iterations. However, in the later stages, UC exhibits greater accuracy gains than PS. These results suggest that UC is more advantageous than PS in scenarios where determining the plausibility of a triple is more challenging. In other words, the uncertainty inherent in model predictions is more

TABLE 3. Link prediction performance on FB15K237. The left and right sides of the table show the results for all triples and results for triples in the minor 50 relations, respectively.

Sampling	All Relations				Minor 50 Relations			
	MRR	Hits@10	Hits@3	Hits@1	MRR	Hits@10	Hits@3	Hits@1
Original (100%)	0.289	0.476	0.327	0.194	0.414	0.611	0.475	0.306
No AL (80%)	0.199	0.380	0.234	0.105	0.308	0.536	0.381	0.188
RD	0.211	0.393	0.245	0.117	0.327	0.561	0.392	0.202
PS	0.257	0.442	0.291	0.164	0.388	0.579	0.447	0.280
PS-balance	0.234	0.420	0.271	0.137	0.390	0.587	0.461	0.284
PS-cluster	0.259	0.444	0.293	0.167	0.391	0.588	0.458	0.281
UC	0.261	0.445	0.298	0.169	0.393	0.582	0.454	0.283
UC-balance	0.232	0.416	0.266	0.137	0.391	0.583	0.459	0.286
UC-cluster	0.263	0.450	0.295	0.170	0.394	0.583	0.455	0.285

effective in identifying whether a triple is correct than the plausibility score provided by the model. As illustrated in Figure 6, UC performs slightly better than PS in the later AL iterations, even on the WikiMovie1M, which contains the highest number of relations and entities.

We also evaluated AL sampling strategies that consider the relation distribution in real-world KGs: balanced and clustered sampling. Figure 7 illustrates the relation distribution in the initial KG, i.e., FB15K237, and the distribution of the triples improved by each sampling strategy. Sampling strategies such as RD, PS, and UC, which disregard the underlying relation distribution, tend to follow the long-tail distribution of the initial KG. In contrast, balanced sampling strategies like PS-balance and UC-balance yield more evenly distributed improvements across all relations. However, as demonstrated in Figures 4 and 5, balanced sampling strategies show slightly lower performance compared to other strategies in terms of absolute KG accuracy improvement. Clustered sampling strategies provide more balanced improvements in KG accuracy compared to RD while exhibiting a distribution pattern similar to normal PS and UC. Notably, the clustered variants, PS-cluster and UC-cluster, achieve significantly greater improvements compared to their non-clustered sampling strategies. These results suggest that clustered sampling enhances relation-specific accuracy by exploiting the semantic similarity of triples within the same cluster.

D. IMPACTS OF SAMPLING STRATEGIES ON DOWNSTREAM TASKS (RQ4)

We conducted link prediction experiments to evaluate the impact of KG accuracy improvement under different AL sampling strategies on downstream tasks. The dataset used in the experiment is FB15K237, and the ratio of incorrect triples is set to 20%. Also, DistMult is employed as the model for the AL framework, while TransE is used as the model for link prediction. The experiments are performed on the OpenKE,² a widely used platform for KG embedding, with hyperparameters set to their default configurations.

The left side of Table 3 shows the link prediction results for all relations in the FB15K237, i.e., all triples. First, when the AL framework was not applied (No AL), the Hit@10

²<https://github.com/thunlp/OpenKE>

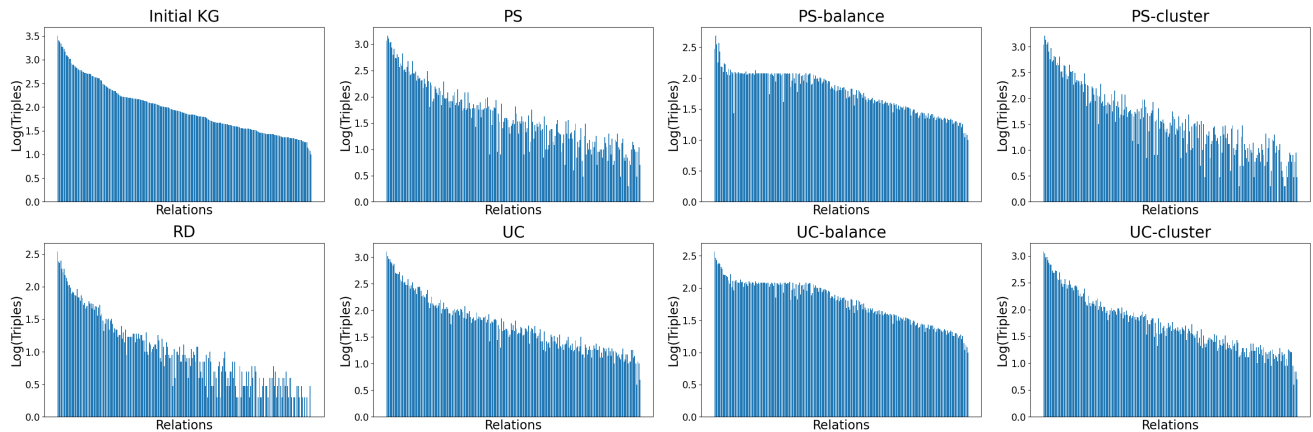


FIGURE 7. Relation distribution in the initial KG and the distribution of the triples improved by each AL sampling strategy. Relations are sorted in descending order based on the number of triples within the initial KG.

value decreases by approximately 0.1 p compared to training with the original KG without incorrect triples (Original). By applying the AL framework, we observed that MRR and Hit@N values increase proportionally to the extent of KG accuracy improvement for each AL sampling strategy. These results imply that the proposed framework improves KG accuracy and enhances downstream tasks' performance.

The right side of Table 3 presents the link prediction results for triples in the minor 50 relations infrequent in the original FB15K237. Similar to the results above, applying the AL framework leads to an increase in both MRR and Hit@N. Notably, sampling strategies considering relation distribution in the KG, such as balanced and clustered sampling, result in remarkable performance improvements. In particular, strategies that applied balanced sampling exhibit the best link prediction performance despite achieving a smaller KG accuracy improvement than other sampling strategies. These results indicate that the proposed distribution-aware sampling strategies effectively enhance the accuracy of triples belonging to minor relations by alleviating the sample imbalance caused by the long-tail distribution in KG.

E. FURTHER ANALYSIS OF SAMPLING STRATEGIES

We conducted additional experiments to investigate AL sampling strategies empirically. First, we analyzed the performance variations with respect to the number of clusters in the clustered sampling strategy. Figure 8 illustrates the relationship between the number of clusters and the improvements in KG accuracy. Empirically, we observed that the optimal number of clusters was greater than or equal to the number of relations within the KG. For both PS- and UC-cluster sampling strategies, improvements in KG accuracy are consistently higher when the number of clusters exceeds the number of relations in the KG and continues to increase with larger numbers of clusters. These results indicate that semantic categorization through clustering is more effective in enhancing performance than explicit categorization of triples based on their relations. Notably, this effect is more

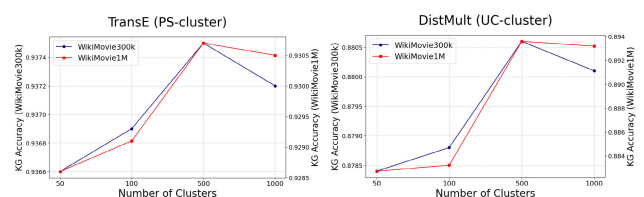


FIGURE 8. Comparison of KG accuracy improvement according to the number of clusters in the clustered sampling.

pronounced in UC compared to PS. This result also suggests that, within semantically clustered triples, selecting triples with high uncertainty yields a more significant sampling performance. Furthermore, the larger accuracy improvements observed in WikiMovie1M compared to WikiMovie300K indicate that clustered sampling strategies are effective for KGs with a greater diversity of relations and entity types.

We also examined the impact of dynamically adjusting the number of clusters on the improvement of KG accuracy. In this experiment, we employed DistMult, setting the default number of clusters to 500. The number of clusters is modified by doubling (up) or halving (down) after reaching the halfway point of the AL iteration. As illustrated in Figure 9, the overall trend in accuracy improvement remains consistent regardless of cluster adjustments. However, downsizing the number of clusters yields slightly better performance than increasing it. Moreover, it is observed that the performance variations with respect to cluster adjustments are relatively minor in the UC sampling compared to the PS sampling. These findings indicate that dynamically adjusting the number of clusters can be an effective strategy for balancing computational efficiency and performance in the proposed framework.

VI. HUMAN-IN-THE-LOOP EXPERIMENTS

This section introduces human-in-the-loop experiments to explore the impact of human involvement. In the proposed AL framework, the involvement of human oracles, responsible for rectifying incorrect triples, plays a critical

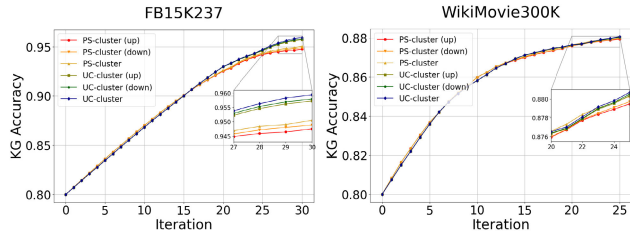


FIGURE 9. Comparison of KG accuracy at each AL iteration with dynamic adjustment of cluster size.

role. The expertise of these oracles significantly affects the improvement in KG accuracy and the overall efficiency of the framework. Notably, their expertise becomes even more crucial when working with domain-specific KGs. Therefore, we investigated the impact of the domain knowledge of human oracles, i.e., the level of expertise, on improving KG accuracy.

A. EXPERTISE LEVEL OF HUMAN ORACLES

We categorized human domain expertise for improving the KG into three levels: Expert, Semi-expert, and Non-expert. A domain expert is someone with specialized knowledge in a particular area or industry. They possess structured expertise, enabling them to deliver reliable solutions grounded in a comprehensive understanding of the domain. A semi-expert refers to an individual with advanced knowledge in a particular field who has not yet attained the status of a specialist with fully comprehensive expertise. While they do not independently apply structured knowledge, they can understand and utilize it effectively when provided with a domain context. Finally, a non-expert is an individual who lacks the specialized knowledge required to perform tasks within a specific domain.

To reflect these levels of expertise in the human-in-the-loop triple rectification process, we designed the following human oracle conditions:

- Non-expert: No contextual or structured knowledge is available to support triple rectification.
- Semi-expert: Textual context is provided for the head, relation, and tail of each triple. Human annotators, acting as semi-experts, perform rectification by referencing the corresponding triple-aligned text.
- Expert: External KG is provided to facilitate triple rectification. Human annotators, acting as experts, are enabled to access the KG exploration tool.

To explore the impact of these expertise levels, we selected UMLS, a KG from the biomedical domain, as the target for evaluation. In the semi-expert condition, the UMLS triple-text alignment dataset provided by Yao et al. [47] is utilized as a reference text. Also, the UMLS Semantic Network Browser³ is employed as an external KG exploration tool in the expert condition.

TABLE 4. Performance of the AL framework for improving KG accuracy according to the level of expertise of human oracles.

Lv. of Expertise	$Acc(\Delta)$	Eff_{anno}	instance/sec
Non-expert	0.8092($\uparrow 0.0092$)	0.0545	-
Semi-expert	0.8553($\uparrow 0.0553$)	0.5214	178.32
Expert	0.8857($\uparrow 0.0857$)	0.8307	41.64

B. ANNOTATORS AND GUIDELINES

We conducted human-in-the-loop experiments by employing three graduate students as annotators.⁴ The annotators were proficient in English and possessed a substantial background in KGs and ontologies, although they lacked expertise in the biomedical domain. The annotators were asked to rectify the incorrect entity in a given triple based on their designated level of expertise within the experimental conditions. For ease of evaluation, we explicitly informed the annotators which entity was incorrect. Furthermore, to prevent duplication of rectified triples, a search function was provided to allow the annotators to explore the current state of the target KG.

C. ADDITIONAL SETTINGS

The AL framework was configured with the following settings: KGE model: TransE; Sampling strategy: PS-cluster; Number of clusters: 100; Sampling size: 100; and AL iterations: 10. Additionally, incorrect triples were generated at a ratio of 0.2, following the process described in IV-A.

Annotation Efficiency (Eff_{anno}) according to the level of expertise of the human oracle was calculated as follows:

$$Eff_{anno} = \frac{\sum_{k=1}^K \frac{|\mathcal{S}_k^{rectified}|}{|\mathcal{S}_k^{incorrect}|}}{K} \quad (11)$$

where $\mathcal{S}_k^{incorrect}$ and $\mathcal{S}_k^{rectified}$ are the incorrect triples and the rectified triples in selected samples $\mathcal{S}_k$ at k -th AL iteration, respectively. The higher value of Eff_{anno} indicates that the human oracle achieves higher annotation accuracy, reflecting an improved annotation quality.

D. IMPACTS OF HUMAN ORACLES ON KG ACCURACY

Table 4 presents the results of KG accuracy improvement based on the expertise level of the human oracle. For non-experts, the accuracy improvement is observed to be 0.92% p, with an annotation efficiency of 0.0545. These findings indicate that the involvement of non-experts is challenging to lead to significant accuracy improvement. When semi-experts serve as human oracles, the accuracy improvement increases to 5.53% p, with an annotation efficiency of 0.5214. The expert achieves a significantly higher accuracy improvement of 8.57% p and an annotation efficiency of 0.8307. Notably, on average, semi-experts required approximately four times the improvement time compared to experts. These results highlight the role of the

³<https://uts.nlm.nih.gov/uts/umls/semantic-network>

⁴They were compensated appropriately by institutional regulations.

human oracle's expertise level in enhancing the performance of the framework. Moreover, they suggest that financial and time costs, as well as overall efficiency, may vary depending on the expertise of the participating oracles. Consequently, selecting human oracles with the appropriate expertise level is crucial for achieving effective and efficient accuracy improvements within a given budget.

VII. DISCUSSIONS

In this section, we discuss the key considerations for applying the AL framework to enhance KG accuracy in real-world scenarios. Also, we examine the some limitations of the proposed framework.

A. COST OF KG ACCURACY IMPROVEMENT

The proposed framework necessitates training an embedding model for the target KG to identify suspicious triples. As previously noted, real-world KGs comprise a large number of triples representing various entities, relations, and their interconnections. Given this scale, training a KGE model can result in substantial time and computational costs. Furthermore, these costs may escalate exponentially depending on the model's complexity and the nature of the input KG. Consequently, it is crucial for the operator to carefully select the most suitable KGE model, taking into account the size and characteristics of the target KG. Furthermore, while not explored in this study, future research could consider the incorporation of incremental learning to mitigate retraining costs, as well as the adoption of a distributed training framework to facilitate the processing of large-scale KGs.

In the proposed framework, the expertise of human oracles, responsible for accurately rectifying incorrect triples, is a critical factor. KGs encompass not only general knowledge, such as commonsense, but also domain expert knowledge. Unfortunately, employing human oracles with domain-specific expertise incurs significantly higher economic costs. Therefore, selecting an optimal sampling strategy that maximizes sampling efficiency is crucial to ensuring the effective utilization of expert labor within limited AL budgets. Moreover, the expertise level of human oracles involved in the AL framework is also an important factor influencing its overall performance. As demonstrated in Section VI, even for relatively small-scale KGs, both the quality of improvements and the time required for refinement vary depending on the oracles' level of expertise. Therefore, in order to improve the KG quality in real-world scenarios, it is essential to account for the expertise level of participants appropriately.

B. GOAL OF KG ACCURACY IMPROVEMENT

The choice of the KGE model and sampling strategy may differ depending on the specific objective for enhancing the KG accuracy. For instance, if the goal is to maximize the overall accuracy of the KG, the model and sampling strategy that achieves the most significant improvement in accuracy

should be selected. Conversely, if the objective is to enhance the accuracy of triples associated with infrequent relations, a model with strong generalization capabilities, coupled with a sampling strategy that considers the distribution of relations, should be employed. Therefore, it is imperative for operators to carefully select the appropriate KGE model and sampling strategy that is in alignment with the desired objectives of KG improvement.

C. LIMITATIONS AND FUTURE WORK

In the proposed framework, only the internal information of the KG is utilized to identify incorrect triples. However, KGs often incorporate auxiliary information, such as textual descriptions and ontological schemas, which can further enhance the accuracy of the KG. Moreover, recent studies on multimodal KGs, which aim to integrate diverse sources of information, have been attracting increasing attention. As part of our future work, we plan to extend the proposed framework to leverage various forms of auxiliary information for comprehensive KG refinement. In addition, the proposed AL framework relies on costly human involvement in the triple rectification process. To mitigate this dependency, we are considering the potential of leveraging reinforcement learning agents or LLMs, which have demonstrated remarkable performance across various human-substitution tasks.

VIII. CONCLUSION

In this paper, we introduced a novel AL framework designed to enhance the accuracy of KGs. The proposed framework efficiently identifies incorrect triples and rectifies them accurately with the involvement of the human oracle. Additionally, we presented sampling strategies that account for the relation distribution in the real-world KG, aiming to maximize sampling efficiency within a limited AL budget. Comprehensive experimental results demonstrate that the proposed AL framework and sampling strategies are effective in improving KG accuracy. Furthermore, we provided practical guidelines for applying AL to KGs based on comprehensive discussions.

REFERENCES

- [1] X. Wang, X. He, Y. Cao, M. Liu, and T.-S. Chua, "KGAT: Knowledge graph attention network for recommendation," in *Proc. 25th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Jul. 2019, pp. 950–958.
- [2] J. Jung, B. Son, and S. Lyu, "AttnIO: Knowledge graph exploration with in-and-out attention flow for knowledge-grounded dialogue," in *Proc. Conf. Empirical Methods Natural Lang. Process. (EMNLP)*, 2020, pp. 3484–3497.
- [3] M. Yasunaga, H. Ren, A. Bosselut, P. Liang, and J. Leskovec, "QA-GNN: Reasoning with language models and knowledge graphs for question answering," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics: Human Lang. Technol.*, 2021, pp. 535–546.
- [4] S. Pan, L. Luo, Y. Wang, C. Chen, J. Wang, and X. Wu, "Unifying large language models and knowledge graphs: A roadmap," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 7, pp. 3580–3599, Jul. 2024.
- [5] X. Shi, Z. Zhu, Z. Zhang, and C. Li, "Hallucination mitigation in natural language generation from large-scale open-domain knowledge graphs," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2023, pp. 12506–12521.

- [6] G. Agrawal, T. Kumarage, Z. Alghamdi, and H. Liu, "Can knowledge graphs reduce hallucinations in LLMs? : A survey," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics, Human Lang. Technol.*, 2024, pp. 3947–3960.
- [7] L. Luo, Y.-F. Li, G. Haffari, and S. Pan, "Reasoning on graphs: Faithful and interpretable large language model reasoning," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Jan. 2023, pp. 1–43.
- [8] J. Sun, C. Xu, L. Tang, S. Wang, C. Lin, Y. Gong, H.-Y. Shum, and J. Guo, "Think-on-graph: Deep and responsible reasoning of large language model with knowledge graph," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2024, pp. 1–14.
- [9] B. Xue and L. Zou, "Knowledge graph quality management: A comprehensive survey," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 5, pp. 4969–4988, May 2023.
- [10] J. A. Carlson, J. Betteridge, B. Kisiel, B. Settles, E. R. Hruschka, and T. M. Mitchell, "Toward an architecture for never-ending language learning," in *Proc. AAAI Conf. Artif. Intell.*, vol. 24, Jul. 2010, pp. 1306–1313.
- [11] F. Akrami, M. S. Saeef, Q. Zhang, W. Hu, and C. Li, "Realistic re-evaluation of knowledge graph completion methods: An experimental study," in *Proc. ACM SIGMOD Int. Conf. Manage. Data*, Jun. 2020, pp. 1995–2010.
- [12] P. Ojha and P. Talukdar, "KGEval: Accuracy estimation of automatically constructed knowledge graphs," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2017, pp. 1741–1750.
- [13] J. Gao, X. Li, Y. E. Xu, B. Sisman, X. L. Dong, and J. Yang, "Efficient knowledge graph accuracy evaluation," *Proc. VLDB Endowment*, vol. 12, no. 11, pp. 1679–1691, Jul. 2019.
- [14] Y. Qi, W. Zheng, L. Hong, and L. Zou, "Evaluating knowledge graph accuracy powered by optimized human-machine collaboration," in *Proc. 28th ACM SIGKDD Conf. Knowl. Discovery Data Mining*, Aug. 2022, pp. 1368–1378.
- [15] S. Jia, Y. Xiang, X. Chen, and K. Wang, "Triple trustworthiness measurement for knowledge graph," in *Proc. World Wide Web Conf.*, May 2019, pp. 2865–2871.
- [16] Y. Wang, F. Ma, and J. Gao, "Efficient knowledge graph validation via cross-graph representation learning," in *Proc. 29th ACM Int. Conf. Inf. Knowl. Manage.*, Oct. 2020, pp. 1595–1604.
- [17] Q. Zhang, J. Dong, K. Duan, X. Huang, Y. Liu, and L. Xu, "Contrastive knowledge graph error detection," in *Proc. 31st ACM Int. Conf. Inf. Knowl. Manage.*, Oct. 2022, pp. 2590–2599.
- [18] J. Dong, Q. Zhang, X. Huang, Q. Tan, D. Zha, and Z. Zihao, "Active ensemble learning for knowledge graph error detection," in *Proc. 16th ACM Int. Conf. Web Search Data Mining*, Feb. 2023, pp. 877–885.
- [19] X. Liu, Y. Liu, and W. Hu, "Knowledge graph error detection with contrastive confidence adaption," in *Proc. AAAI Conf. Artif. Intell.*, vol. 38, Mar. 2024, pp. 8824–8831.
- [20] A. Kirsch, J. V. Amersfoort, and Y. Gal, "BatchBALD: Efficient and diverse batch acquisition for deep Bayesian active learning," in *Proc. Int. Conf. Neural Inf. Process. Syst. (NeurIPS)*, Jan. 2019, pp. 1–54.
- [21] D. Yoo and I. S. Kweon, "Learning loss for active learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 93–102.
- [22] O. Sener and S. Savarese, "Active learning for convolutional neural networks: A core-set approach," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Jan. 2017, pp. 1–54.
- [23] T. He, X. Jin, G. Ding, L. Yi, and C. Yan, "Towards better uncertainty sampling: Active learning with multiple views for deep convolutional neural network," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, Jul. 2019, pp. 1360–1365.
- [24] J. T. Ash, C. Zhang, A. Krishnamurthy, J. Langford, and A. Agarwal, "Deep batch active learning by diverse, uncertain gradient lower bounds," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Jan. 2019, pp. 1–47.
- [25] M. Yuan, H.-T. Lin, and J. Boyd-Graber, "Cold-start active learning through self-supervised language modeling," in *Proc. Conf. Empirical Methods Natural Lang. Process. (EMNLP)*, 2020, pp. 7935–7948.
- [26] N. Ostapuk, J. Yang, and P. Cudre-Mauroux, "ActiveLink: Deep active learning for link prediction in knowledge graphs," in *Proc. World Wide Web Conf.*, May 2019, pp. 1398–1408.
- [27] A. Pradhan, K. K. Todi, A. Selvarasu, and A. Sanyal, "Knowledge graph generation with deep active learning," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jul. 2020, pp. 1–8.
- [28] B. Liu, H. Scells, G. Zuccon, W. Hua, and G. Zhao, "ActiveEA: Active learning for neural entity alignment," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2021, pp. 3364–3374.
- [29] S. Seo, B. Oh, E. Jo, S. Lee, D. Lee, K.-H. Lee, D. Shin, and Y. Lee, "Active learning for knowledge graph schema expansion," *IEEE Trans. Knowl. Data Eng.*, vol. 34, no. 12, pp. 5610–5620, Dec. 2022.
- [30] J. Huang, Z. Sun, Q. Chen, X. Xu, W. Ren, and W. Hu, "Deep active alignment of knowledge graph entities and schemata," *Proc. ACM Manage. Data*, vol. 1, no. 2, pp. 1–26, Jun. 2023.
- [31] A. Bordes, N. Usunier, A. García-Durán, J. Weston, and O. Yakhnenko, "Translating embeddings for modeling multi-relational data," in *Proc. Int. Conf. Neural Inf. Process. Syst. (NeurIPS)*, Dec. 2013, pp. 1–23.
- [32] T. Wu, A. Khan, M. Yong, G. Qi, and M. Wang, "Efficiently embedding dynamic knowledge graphs," *Knowl.-Based Syst.*, vol. 250, Aug. 2022, Art. no. 109124.
- [33] Y. Mao, Y. Hao, X. Cao, Y. Fang, X. Lin, H. Mao, and Z. Xu, "Dynamic graph embedding via meta-learning," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 7, pp. 2967–2979, Jul. 2024.
- [34] W. Xiong, M. Yu, S. Chang, X. Guo, and W. Y. Wang, "One-shot relational learning for knowledge graphs," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2018, pp. 1980–1990.
- [35] D. Arthur and S. Vassilvitskii, "K-means++: The advantages of careful seeding," in *Proc. 18th Annu. ACM-SIAM Symp. Discrete Algorithms*, Jan. 2007, pp. 1027–1035.
- [36] K. Toutanova, D. Chen, P. Pantel, H. Poon, P. Choudhury, and M. Gamon, "Representing text for joint embedding of text and knowledge bases," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2015, pp. 1499–1509.
- [37] T. Dettmers, P. Minervini, P. Stenetorp, and S. Riedel, "Convolutional 2D knowledge graph embeddings," in *Proc. AAAI Conf. Artif. Intell.*, vol. 32, Apr. 2018, pp. 1811–1818.
- [38] R. Xie, Z. Liu, F. Lin, and L. Lin, "Does William Shakespeare REALLY write hamlet? Knowledge representation learning with confidence," in *Proc. AAAI Conf. Artif. Intell.*, Jan. 2017, pp. 4954–4961.
- [39] T. Ebisu and R. Ichise, "ToruSE: Knowledge graph embedding on a lie group," in *Proc. AAAI Conf. Artif. Intell.*, vol. 32, Apr. 2018, pp. 1819–1826.
- [40] B. Yang, W.-T. Yih, X. He, J. Gao, and L. Deng, "Embedding entities and relations for learning and inference in knowledge bases," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Jan. 2014, pp. 1–12.
- [41] T. Trouillon, J. Welbl, S. Riedel, É. Gaussier, and G. Bouchard, "Embedding entities and relations for learning and inference in knowledge bases," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2016, pp. 2071–2080.
- [42] J. Shen, C. Wang, Y. Yuan, J. Han, H. Ji, K. Sen, M. Zhang, and D. Song, "PALT: Parameter-lite transfer of language models for knowledge graph completion," in *Proc. Findings Assoc. Comput. Linguistics*, 2022, pp. 3833–3847.
- [43] A. Boschin, "TorchKGE: Knowledge graph embedding in Python and PyTorch," 2020, *arXiv:2009.02963*.
- [44] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.
- [45] J. Devlin, M. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics, Hum. Lang. Technol. (NAACL-HLT)*, Jan. 2018, pp. 4171–4186.
- [46] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Jan. 2017, pp. 1–45.
- [47] L. Yao, C. Mao, and Y. Luo, "KG-BERT: BERT for knowledge graph completion," 2019, *arXiv:1909.03193*.



DONGHYUN KIM received the B.S. degree in computer science and engineering from Pusan National University, Republic of Korea, in 2017. He is currently pursuing the Ph.D. degree with the Department of Computer Science, Yonsei University. His research interests include knowledge graphs, natural language processing, and conversational AI.



HYEONGJUN YANG received the B.S. degree in software engineering from Dankook University, Republic of Korea, in 2020. He is currently pursuing the Ph.D. degree with the Department of Computer Science, Yonsei University. His research interests include knowledge-enhanced recommendation and conversational AI.



MIDAN SHIM received the B.S. degree in biology and software engineering from Kyung Hee University, Republic of Korea, in 2022. She is currently pursuing the Ph.D. degree with the Department of Computer Science, Yonsei University. Her research interests include knowledge-based question answering and document AI.



SEOKJU HWANG received the B.S. degree in computer science and engineering from the Tech University of Korea, Republic of Korea, in 2022. He is currently pursuing the Ph.D. degree with the Department of Computer Science, Yonsei University. His research interests include emotional support conversation and goal-oriented dialogue.



KYUHWAN YEOM received the B.S. degree in computer science from Yonsei University, Republic of Korea, in 2022, where he is currently pursuing the Ph.D. degree with the Department of Computer Science. His research interests include knowledge representation learning and knowledge graph completion.



KYONG-HO LEE (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in computer science from Yonsei University, Republic of Korea, in 1995, 1997, and 2001, respectively. He is currently a Professor with the Department of Computer Science, Yonsei University. Previously, he was a Guest Researcher with the IT Laboratory, National Institute of Standards and Technology (NIST), USA. He was also a Visiting Scholar with the University of California at Irvine, Irvine, CA, USA, in 2008. His research interests include knowledge graphs and the semantic web.

...